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A Case Study on Reduced Cluster Spacing: Integrated Modeling and Field Diagnostics from a Permian Trial

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Abstract

This study investigates the impact of reducing cluster spacing on fracture propagation, cluster efficiency, and well productivity in an unconventional Permian Basin pilot. The five-bench field trialat a density of 20 wells per section was comprised of two pads: one with cluster half-spacing (0.5X) and a control at standard cluster spacing (1X). We employed an integrated workflow combining analytical modeling, full-scale physics-based simulation, and multi-modality diagnostics including fiber optics, time-lapsed geochemistry, bottom-hole pressure monitoring, and production history. The numerical model was first calibrated against the half-spacing pad data, then validated without further tuningon the control pad. Its blind predictions of cumulative oil closely matched observed production, demonstrating the approach's robustness. Finally, the calibrated model provided quantitative insight into the efficiency gains achieved by halving cluster spacing.

Introduction

Unconventional reservoirs are characterized by ultra-low matrix permeability often in the nano to micro-Darcy range making hydraulic fracturing essential for economic hydrocarbon recovery. In such tight rocks, well productivity hinges almost entirely on the total fracture surface area created, since greater surface area provides the flow pathways that connect the reservoir to the wellbore (Wattenbarger 1998).

At first glance, it may seem intuitive that simply adding more fractures along a horizontal lateral will always improve well productivity. As operators continue to reduce cluster spacing (Barba and Villarreal 2023), the question emerges: is there a threshold below which the incremental increase in stimulated reservoir volume (SRV) no longer justifies the additional completion cost? In other words, beyond this point, the pursuit of "more fractures" may begin to erode and in some cases reverse economic return.

In the early times of unconventional shale plays development, the perforation cluster spacing was as wide as 700-900 ft in Barnett and Bakken plays (Xiong 2020). Ever since then, operators have increased the number of perforation clusters per lateral length aimed at maximizing the fracture surface area (Zhang and Fan 2025). Each incremental decrease in cluster spacing within the Midland Basin resulted in improved well productivity while maintaining fracture containment near the wellbore (Jaripatke *etal.* 2018). While tighter cluster spacing may deliver short-term production gains, it remains uncertain whether it yields better long-term performance (Esmaili *etal.* 2021).

Furthermore, variations in matrix permeability and reservoir quality can dramatically shift where that threshold lies. Fowler *etal.* (2019) conducted sensitivity studies to evaluate the impact of matrix permeability on cluster spacing. The results indicated that as matrix permeability increases, the NPV optimal cluster spacing also increases. In low-permeability models, pressure depletion remains localized along the relatively long hydraulic fractures, making production highly sensitive to cluster spacing. Conversely, in high-permeability formations, fluid can travel more freely through the matrix, reducing reliance on hydraulic fractures for connectivity. Similarly, Ibrahim (2024) suggested that the economically optimal cluster spacing was 44 ft for a matrix permeability of 100 nD, whereas a significantly tighter spacing of 18 feet was optimal when the permeability in the model was assumed to be 10 nD. Therefore, the optimal number and spacing of fractures must be tailored not only to operational and economic constraints but to specific reservoir properties.

In practice, implementing tighter cluster spacing while holding proppant intensity (lbs/ft) and fluid intensity (bbls/ft) constant means each cluster receives less total fluid and proppant, typically yielding smaller individual fractures. In multi-bench developments such as those in the Midland Basin, where interference and overlap between fractures across benches are inevitable (Ge *et al.* 2022; Chu *et al.* 2018), closer fracture spacing can be leveraged to manage both intra- and inter-bench communication. However, if clusters are placed too closely, intra-stage stress shadows can inhibit the fracture propagation (Warpinski and Branagan 1989; Cheng 2012), reducing cluster uniformity (Cipolla *et al.* 2024) and limiting how many fractures extend into the far field (Peirce and Bunger 2015). Modeling work by Singh *et al.* (2020) demonstrated that stress shadowing causes fractures to propagate asymmetrically and with nonuniform geometry. They also highlighted that tighter cluster spacing may promote out-of-zone fracture growth. Furthermore, they emphasized the importance of interplay between stress layering and stress shadow effects, which determines the fracture footprint and proppant placement. That being said, Li *et al.* (2025) observed that although tighter cluster spacing is associated with greater in-stage stress shadowing, permanent fiber datasets did not provide evidence that a larger magnitude of stress shadowing always leads to a decrease in cluster uniformity. Instead, their analysis indicated that increasing proppant tonnage per stage had a more pronounced negative impact on cluster uniformity.

In parent–child environments where smaller fracture geometry is desirable to mitigate interference, denser cluster spacing with reduced water and sand per cluster can effectively minimize parent–child interaction, assuming high cluster uniformity is achieved. Conversely, if maintaining fracture dimensions comparable to those achieved with wider spacing is preferred, total fluid and proppant volumes per cluster can be increased to recreate the desired fracture geometry. When validated, these strategies offer multiple benefits: higher fracture counts for increased productivity, the ability to increase well density, improved economics, and a versatile toolkit for managing fracture interference and parent–child effects.

To address these questions at scale, a dedicated cluster spacing program was launched across the ExxonMobil acreage in the Midland sub-basin of the Permian Basin, Texas, USA. Several dozen pads spanning a spectrum of reservoir qualities and bench configurations were selected. Each pad was completed with both baseline and half-spacing cluster designs, enabling a systematic comparison between standard and tighter spacing. This multi-pad, statistically robust rollout quantifies how reservoir permeability, pad density, and stress interaction effects govern the performance and potential trade-offs of tighter cluster spacing under real world conditions.

In this study, we focus on one of these areas to demonstrate an integrated modeling approach that combines analytical workflows, full scale physics-based simulation, and multi-modality diagnostics. In this work, we set out to address two key hypotheses:

- Fracture count scaling: Can halving cluster spacing effectively double the number of both near wellbore and far field propagating fractures, or do stress shadows limit far field growth?
- Productivity gains: Does half spacing deliver accelerated early time rates, improved ultimate recovery, or both, when compared to the standard design?

Methods and workflow

Field trial design

The field trial for this study focused on four pads PAD1, PAD2, PAD3, and PAD4 in the Midland sub-basin of the Permian Basin, Texas. All wells targeted five stacked benches at a consistent well density throughout the section (Fig. 1).

- **Parent pads:**
  - **PAD1 (Half-spacing):** Completed with half the standard cluster spacing.
  - **PAD2 (Control):** Completed with the standard cluster spacing.
- **Child pads:**
  - **PAD3 (Child of PAD2):** Completed with half-spacing to evaluate mitigation of parent–child fracture interference.
  - **PAD4 (Child of PAD1):** Also completed with half-spacing under identical design conditions for reference.

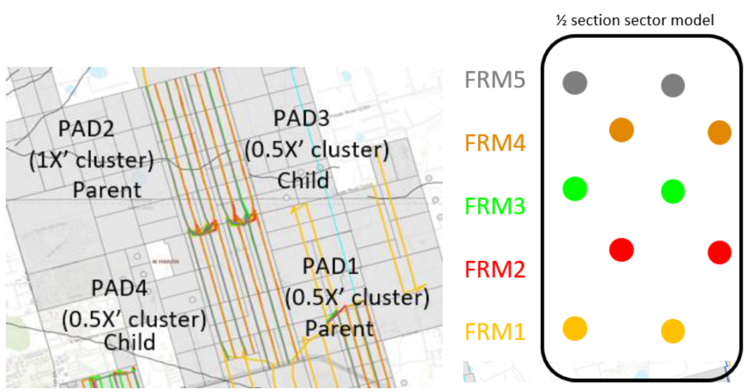


Figure 1—Schematic of pads of interest in Midland Basin

All completions maintained identical proppant and fluid intensities, treatment schedules, and pump rates except in bench FRM4, where intensity was slightly reduced across all pads. These paired pad data underpin the analytical, diagnostic, and integrated modeling workflows presented herein. Modeling efforts are focused on the parent pads PAD1 and PAD2 due to insufficient production history on the child pads; analysis of the child-pad data will be addressed in future work.

Available data and field diagnostics

To build an integrated hydraulic fracture model, several fracture diagnostics, geological and geomechanical datasets are required to constrain the range of free parameters and ensure model validity.

A vertical "type well"near PAD1, was first characterized with a comprehensive log suite including quad-combo sonic, density, and neutron porosityto establish baseline rock properties. On PAD4 (half-spacing), temporary fiber-optic cables in four monitoring wells (two in FRM3 and two in FRM5) captured stage-by-stage Volume-to-First-Response (VFR) and Time-to-First-Response (TFR) signatures, enabling tuning of wetted fracture geometry in the model. Time lapse geochemistry (TLG) data from PAD1 (half-spacing) and PAD2 (standard spacing) and a standalone FRM1 well (south of the pads of interest) were not available during the initial model build but became accessible later allowing independent validation of oil allocation among the benches. Downhole pressure gauges monitored bottomhole pressure, and oil, gas, and water rates over one year were available.

Physics-based fracture and reservoir modeling workflow

For full scale simulations, a fully coupled, physics-based fracturing and reservoir simulator was employed. This integrated package models multistage fracture propagation, proppant transport, fluid leak off, production flowback, and reservoir performance in a single framework.

Static model construction

The static earth model began by integrating wireline and core data to map each layer's petrophysical properties. Quad-combo sonic, density, and neutron porosity logs feed into core–log regressions that yield high resolution porosity, clay volume, and total organic carbon (TOC) profiles. Dipole sonic and density measurements then provide dynamic Young's modulus and Poisson's ratio at every depth. These dynamic values are converted to static equivalents using laboratory triaxial tests, ensuring the model reflects rock deformation under real in-situ stresses.

Pore pressure is built from an extended Eaton normal-compaction trend and then fine-tuned to match the initial gauge pressure measured in every well in the PAD1. After deriving Young's modulus and Poisson's ratio from the logs and converting them to static values, these properties, along with the calibrated pore pressure, are fed into the stress model to compute the minimum principal stress. Since diagnostic fracture injection test (DFIT) measurements were not available, the instantaneous shut-in pressure (ISIP) from PAD1's first stage was used as a proxy to tune those stresses, establishing a consistent framework for all subsequent physics-based fracture simulations. Fig. 2 illustrates some of the geomechanical properties constructed for the vertical well of interest.

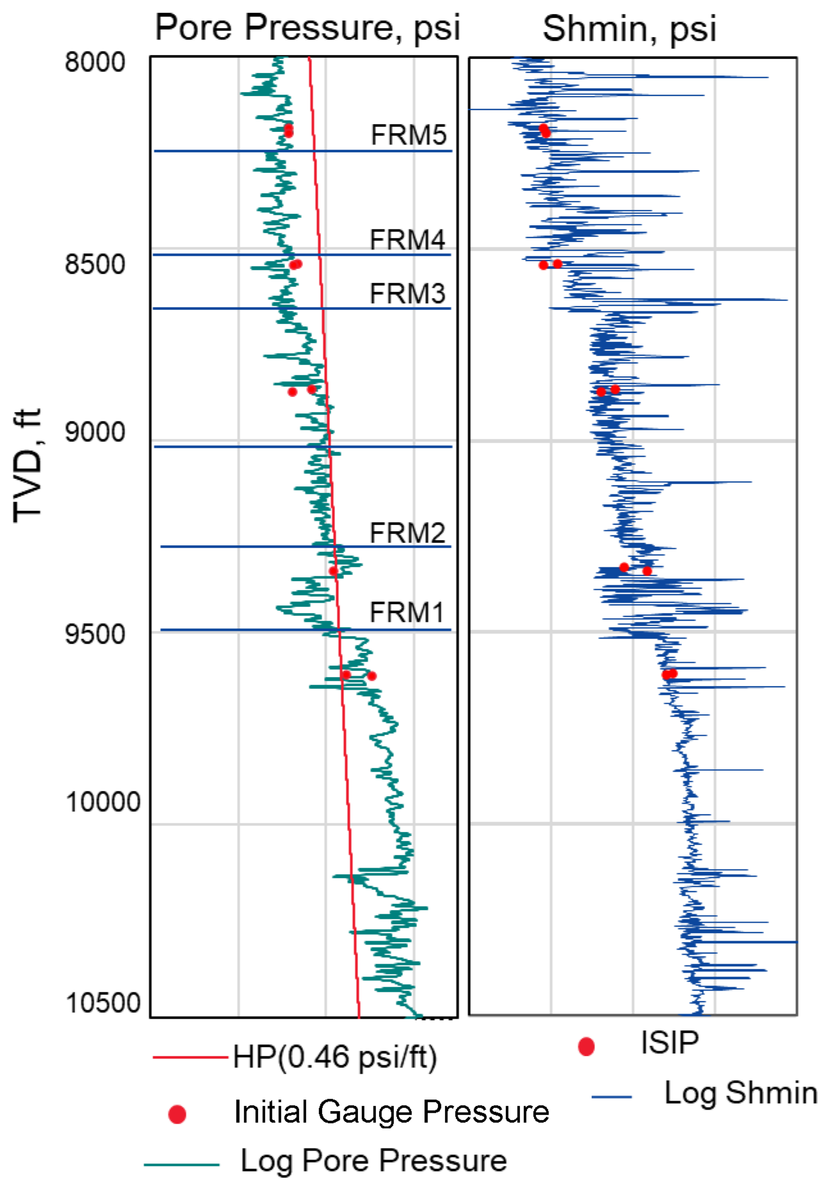


Figure 2—Geomechanics logs constructed for "type well" of interest

Model calibration and history matching

Model calibration follows three sequential steps. First, wetted fracture geometry is tuned using fiber optic data matching simulated geometry to observed TFR and VFR signatures. Second, propped fracture geometry leverages this calibrated wetted geometry and is validated against rate transient analysis (RTA) and TLG data to ensure accurate proppant distribution. Third, production and pressure history matching align simulated oil, gas, and water rates and bottomhole pressure with the one year production record from PAD1, with the two primary tuning parameters being pressure-dependent permeability and relative permeability curves. This structured, multi-dataset workflow ensures each component of the fracture model is independently validated before proceeding to the next step.

Wetted fracture geometry

For the reduced cluster spacing pad, we leveraged a temporary fiber optic system in four monitor wells to capture fracture arrival times; the gun barrel view of PAD4 is illustrated in Fig.3. These arrival times served as our primary calibration target for wetted fracture geometry.

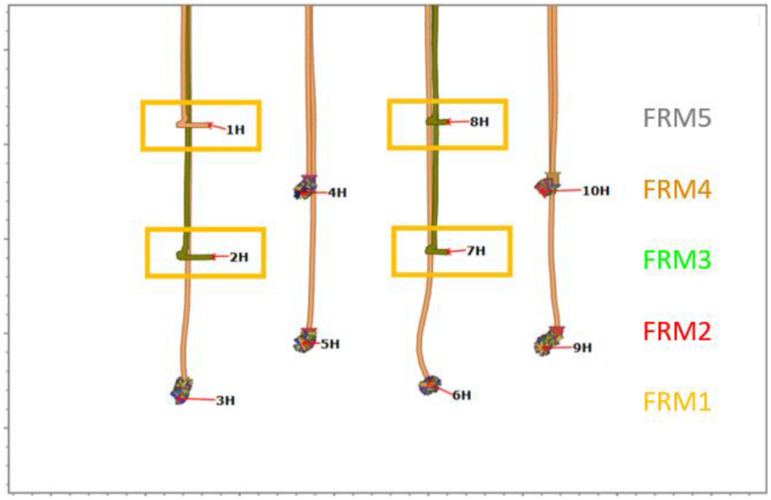


Figure 3—Gunbarrel of PAD4. Wells with fiber cables are highlighted in yellow boxes

To tune the model, a suite of in-situ stress realizations was generated reflecting uncertainty in in-situ stress magnitudes, particularly across potential baffle intervals. Forward fracture simulations were then run for each case. By comparing simulated arrival times at the monitor wells against the fiber-recorded first-response times, we identified the stress realization that best matched the field data. The fracture dimensions from this realization then defined the wetted geometry, ensuring that simulated fractures and proppant placement correspond closely to in-situ behavior.

Fig. 4 shows the final calibrated wetted geometries for benches FRM4, FRM1, and FRM2, along with model-predicted VFR compared to fiber measurements, demonstrating close agreement. Notably, fractures in FRM4 and FRM1 predominantly extend laterally and appear vertically containeddue to a baffle interval that impeded upward growth, whereas fractures in FRM2 exhibit less vertical constraint, allowing both lateral and upward propagation.

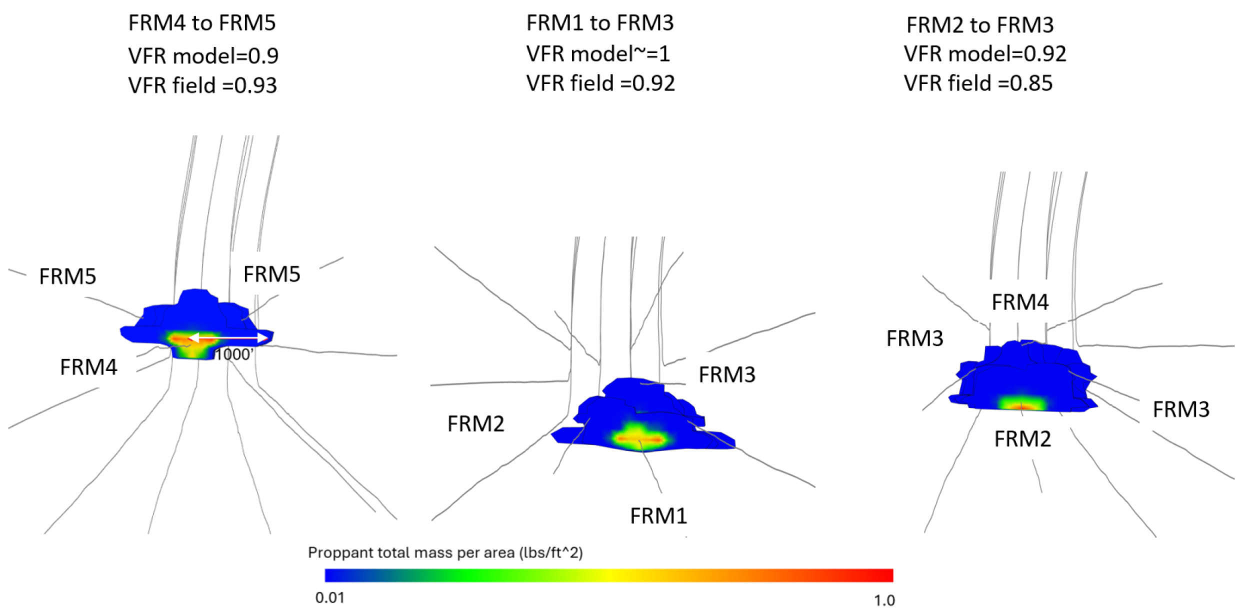


Figure 4—Final calibrated wetted geometries for benches FRM4, FRM1, and FRM2, along with model-predicted VFR compared to fiber measurements

Fiber data captured only the leading arrival of each fracture, so to assess full-stage behavior we used the simulator to propagate all clusters simultaneously. This lets us evaluate cluster uniformity: not just the first fracture arrival but the entire stage.

Model results indicate remarkably high fracture uniformity in both the near field and far field (approaching 100 %), with only minor deviations at distance. Fig. 5 illustrates a stage-level comparison of fractures under standard and half-spacing designs, alongside the VFR curves from FRM1 fractures as recorded at FRM3 and normalized by cluster count. After normalization, the half-spacing design shows only slightly faster arrival times and minimal non-uniformity.

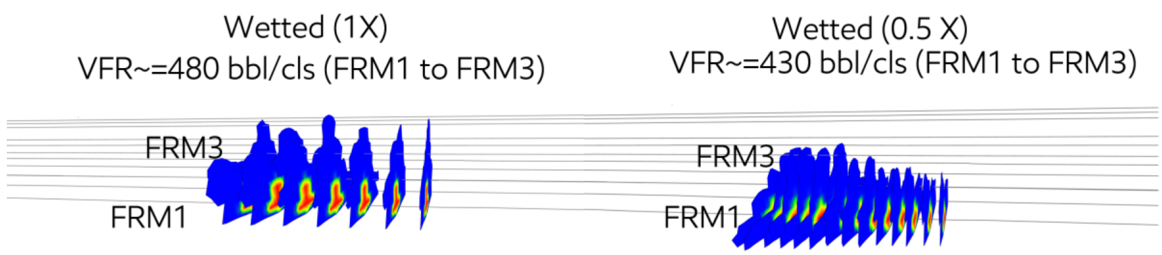


Figure 5—Stage-level comparison of fractures under standard and half-spacing designs, alongside the VFR curves from FRM1 fractures

To confirm these results, two validation approaches were used. First, analytical modeling based on Sneddon's solution (Sneddon 1946) for an elliptical crack in an elastic medium was employed to evaluate stress shadow differences between standard and tighter cluster spacing. Second, near-field diagnostics, specifically in-well fiber optic data and perforation erosion measurements from a nearby pad, provided direct observations of near-wellbore cluster uniformity and efficiency. The following sections detail each of these validation methods.

Analytical modeling

In the analytical modeling phase, the core question was: how much additional stress shadowing arises when cluster spacing is halved? Although simplified, analytical models offer a rapid means to test hypotheses and guide both pilot design and detailed simulations. Sneddon's solution for an elliptical pressurized crack in an elastic medium was used to quantify the stress-shadow effects from adjacent clusters and calculate the change in minimum principal stress ($\Delta\sigma_{min}$) as a function of normalized spacing ($d / \bar{R}$, where d is the distance between clusters and $\bar{R}$ is the average fracture half-length).

The analytical results shown in the top panel of Fig. 6 exhibits very little difference in stress shadowing between 0.5X and 1X cluster spacing. In the bottom panel of Fig. 6, the analytical model shows that in a toughness-dominated regime where fracture net pressure drives fracture growth more than viscous effects, the peak stress shadow at reduced cluster spacing is only about 5 % higher than at standard design. This small increase indicates that the extra stress interaction from tighter spacing is minimal rather than prohibitive, supporting our hypothesis that half-spacing is viable.

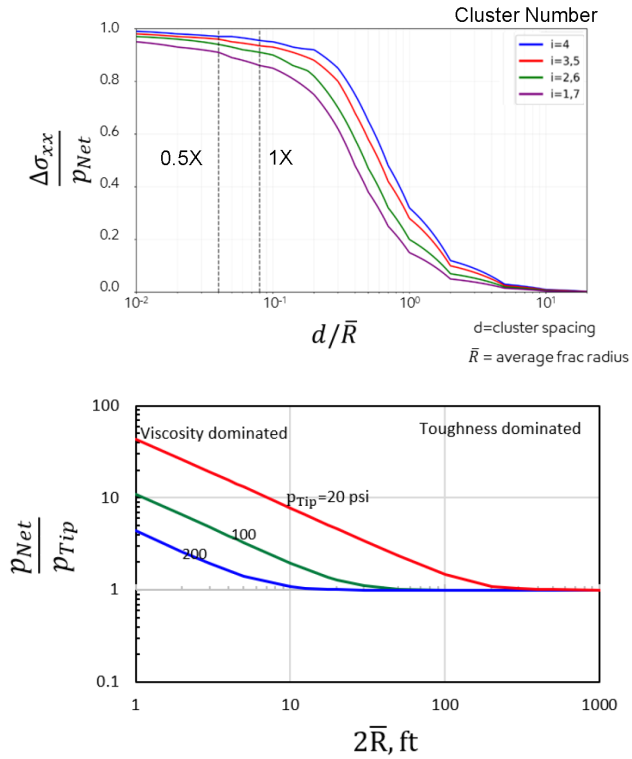


Figure 6—Analytical model for stress shadow measurement as a function of-normalized spacing (d / $\bar{R}$, where $\bar{R}$ is the average fracture half-length).

Near-field cluster uniformity

On a separate pad in the Midland Basin distinct from our primary trial, an advanced near wellbore diagnostic suite to evaluate cluster uniformity under reduced spacing versus standard design was deployed. The schematic gun barrel layout of this pad is illustrated in Fig. 7. Multiple benches were assessed for uniformity index (UI) and cluster efficiency in both reduced spacing and standard designs. The "near-field" pad was instrumented with permanent fiber-optic cables and perforation-erosion logging tools, providing continuous, high-resolution measurements of fluid allocation, uniformity, and near-field cluster efficiency in every stage.

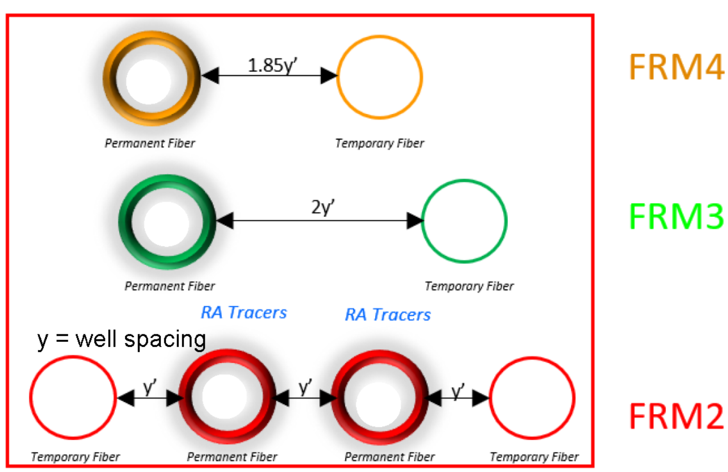
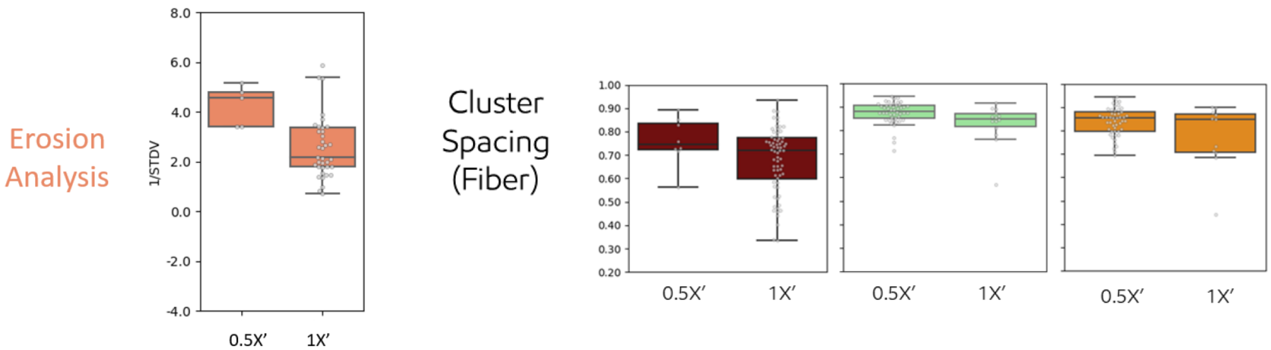


Figure 7—Schematic gun-barrel layout of the pad to measure near field cluster efficiency

The reduced cluster spacing exhibited slightly higher uniformity and overall cluster efficiency compared to the standard design as illustrated in Fig.8, a surprising result that nonetheless agrees with both the simulator's predictions and our analytical modeling. Interestingly, this observation was also seen by Li et al. (2025), as noted earlier. Far-field behavior was not directly measured on this pad, and the trial spacing program lacks data to show how these near-wellbore gains carry into the reservoir at larger scales. Based on both the model and analytical insights, we expect similarly high uniformity in the far field, but actual measurement of far-field fracture performance will be necessary in future work to confirm this prediction.



(a) (b)

Figure 8—Cluster efficiency measurements by (a) erosion log analysis and (b) near-wellbore permanent fiber measurements.

Propped fracture geometry

Propped fracture geometry was defined by the calibrated wetted geometry and the simulator's proppant-transport model, which predicts where proppant settles within each fracture. The simulated propped areas for each bench were checked against facture surface area estimates from rate transient analysis to confirm consistency with production data.

The most compelling validation came from TLG in the standalone FRM1 well: after history matching, the model's bench-by-bench oil-allocation fractions closely matched the FRM1 geochemistry measurements (see Fig. 9).

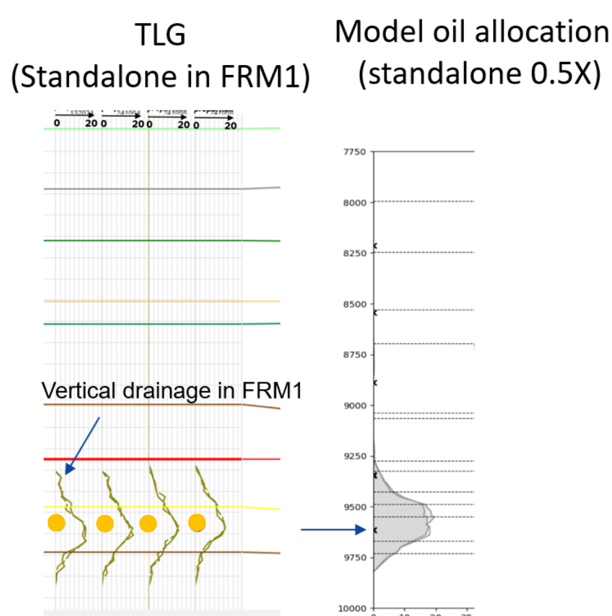


Figure 9—TLG in standalone FRM1 compared to model results

While full-stack TLG can serve as a directional check, it remains a poor proxy for individual propped geometries due to high inter-bench interference. Nonetheless, the model's oil-allocation predictions for full-stack pads remained directionally consistent with the geochemical data.

Production history matching

Production history matching was conducted over a one-year period using a bottom hole pressure-controlled scheme.

Fig.10 illustrates the predicted oil, gas, and water rates from the model (solid line) compared against actual field data (dashed line) and adjusted to ensure a close match.

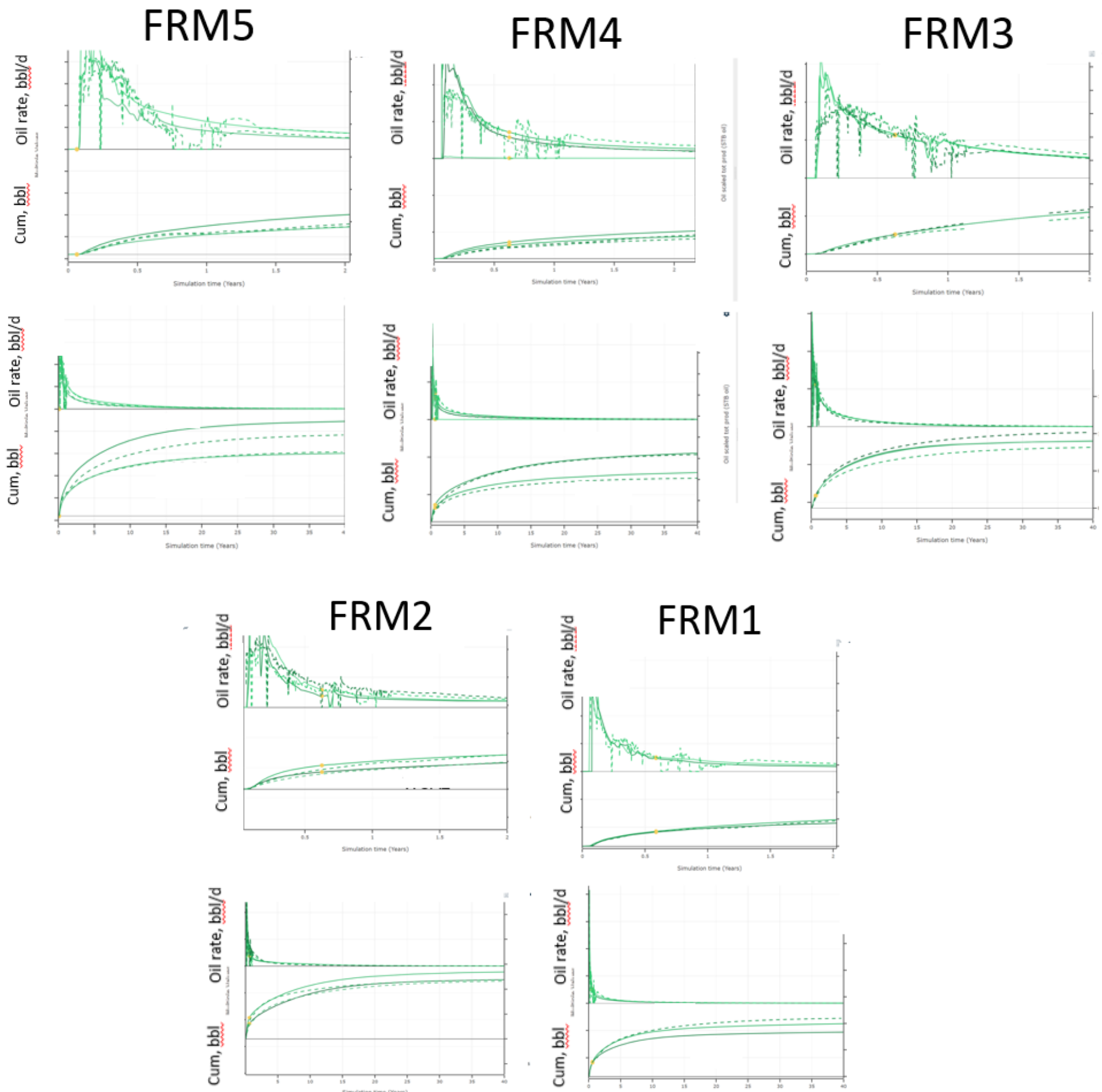


Figure 10—Comparison of predicted oil rates and cumulative production from the model (solid lines) against actual field data (dotted lines)

In early time, a pressure-dependent permeability model captured matrix permeability degradation as reservoir pressure declined, reproducing both the rapid post-fracture rampup and subsequent decline (top panels for each bench: rates and cumulative oil). For the long-term tail, a time-dependent fracture conductivity degradation (Pearson *etal.*, 2022) term was applied via decline-curve analysis to align late-stage rates and estimated ultimate recovery (EUR) (see bottom panels for each bench).

The calibrated model also matched individual well performance across the pad.

Results

Blind test validation

A blind test offers the strongest check on a model's predictive power by confronting it with data to which it has not been calibrated. After history matching PAD1 (reduced spacing), the calibrated model without any further tuning was applied to PAD2 (standard design). Its performance forecasts were then compared directly against PAD2 actual production history, as illustrated in Fig.11. The model successfully predicted the relative performance of each bench, with only minor deviations between forecast and field data.

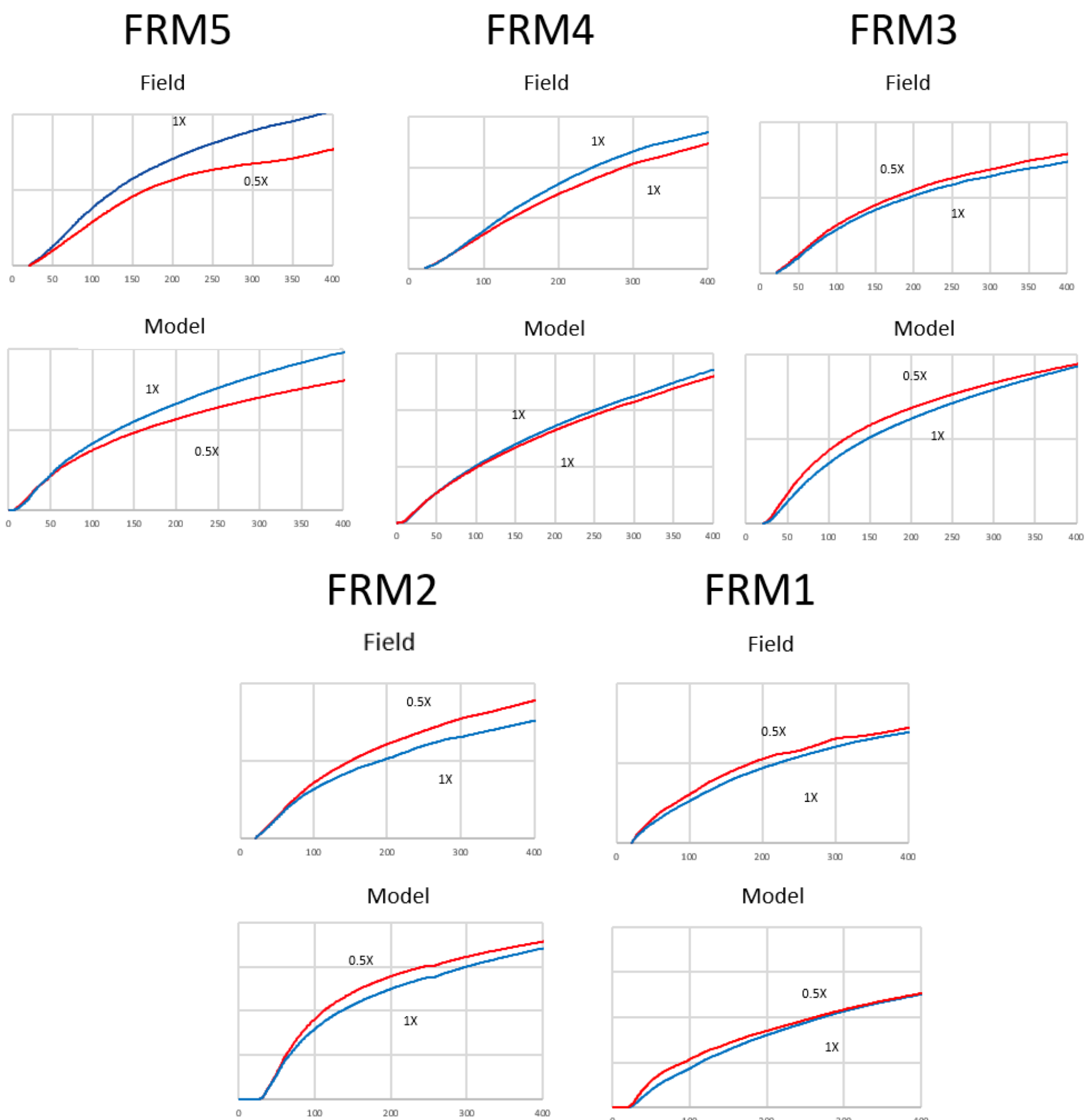


Figure 11—Comparison of performance forecasts compared directly against PAD2's actual production history

Fig. 12 presents a collapsed view of proppant concentration across all modeled fractures for the full stack development under both standard and tighter cluster spacing designs. A threshold of 0.01 lb/ft² was applied as a typical cutoff to define the propped geometry.

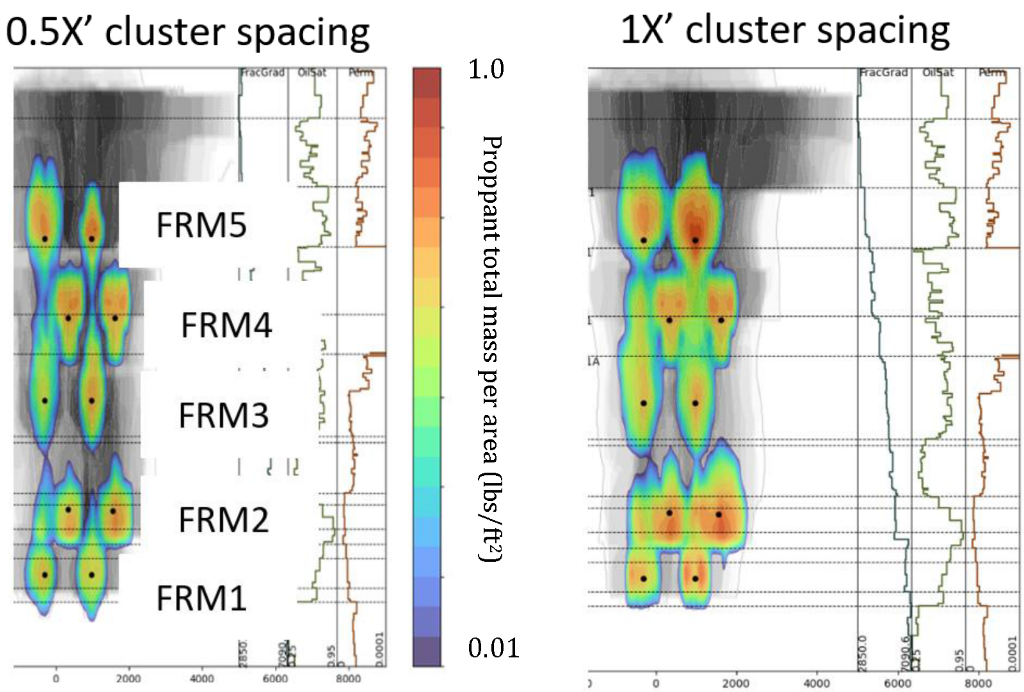


Figure 12—Comparison of propped fracture geometry under standard and reduced cluster-spacing designs

When examining the propped fracture geometries, it becomes clear that, with equal fluid and proppant intensities, the standard-spacing design generally creates a larger individual propped area than the tighter cluster spacing. At first glance, one might assume that bigger fractures are always preferable. This is true when considering a stand alone stage in isolation. However, in a full stack completion, substantial overlap between fractures means that simply maximizing individual fracture size can be counterproductive.

To assess how reduced cluster spacing performs in the trial, it is important to move beyond an individual bench perspective. In the Midland Basin, significant overlap between fractures in adjacent benches makes bench-by-bench evaluation unrealistic. Instead, the concept of "flow units" better captures reservoir behavior. Based on the propped fracture geometries (Fig. 11), two primary flow units emerge: a deeper unit comprising FRM1 and FRM2, and a shallower unit made up of FRM3, FRM4, and FRM5.

Discussion

Based on the flow unit framework and the propped fracture geometries shown in Fig. 12, the trial results reveal a clear pattern which will be discussed in the next two sections:

- In the deeper flow unit (FRM1 + FRM2), the reduced cluster spacing outperforms the standard spacing design. Both field data and the model show higher cumulative oil under the reduced cluster spacing.
- In the shallower flow unit (FRM3 + FRM4 + FRM5), the standard design yields higher recovery.

Deep flow unit

To understand why the reduced cluster spacing design for the FRM1 + FRM2 flow unit outperforms, we ran two simulation tests on a simplified pad containing only these two benches. In the first test, each bench was modeled separately: once with standard spacing and once with reduced cluster spacing so there was no overlap. After one year, both scenarios produced almost the same cumulative oil (see Fig. 13).

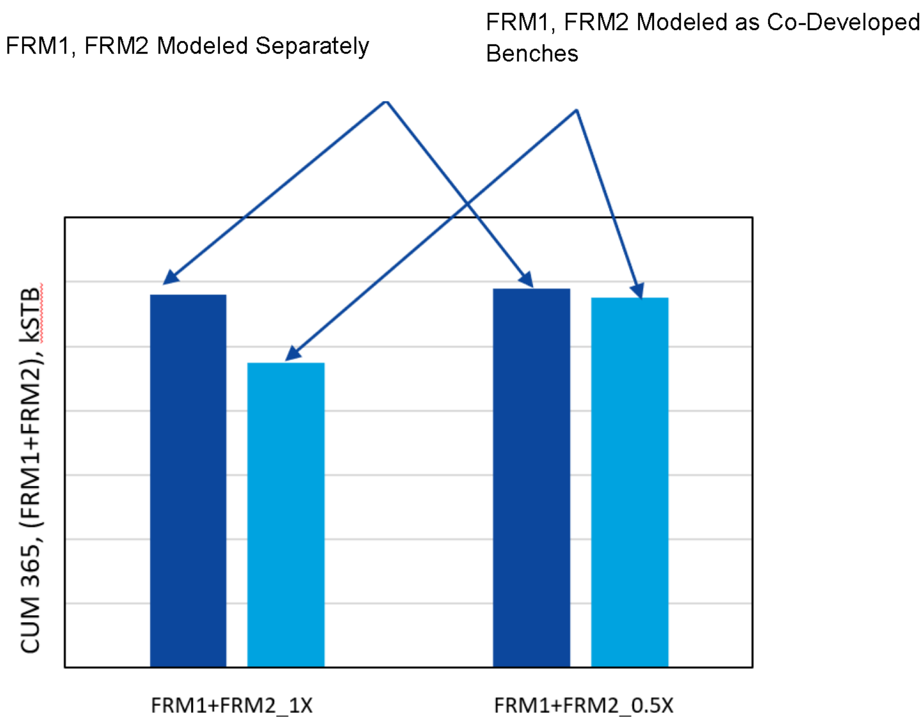


Figure 13—Comparison of one year cumulative production (CUM 365) in standalone vs co-development scheme for deep flow unit

In the second test, both benches were stimulated together (co-developed), allowing overlap between benches. Here, the reduced cluster spacing design delivered higher production. By creating more and smaller fractures, the design reduced the overlap between benches, resulting in better overall performance.

Shallow flow unit

The shallow flow unit (FRM3 + FRM4 + FRM5) exhibits distinct behavior due to variations in reservoir properties. Most notably, FRM4 has permeability an order of magnitude higher than the deeper benches. This elevated permeability means that once a fracture connects to these zones, production becomes dominated by flow through the higher permeability rock.

To isolate the impact of cluster spacing, we performed a detailed two-step simulation comparing a standalone FRM3 bench to a full-stack configuration. Under standard spacing (1X), the model generated tall, highly conductive fractures (Fig. 14, left panel) that penetrated deeply into the FRM3 formation, resulting in approximately 25% higher oil production over the first 365 days compared to the half-spacing (0.5X) scenario (Fig. 15).

In contrast, the 0.5X design produced twice as many fractures, increasing the total stimulated surface area by roughly 30%. However, these fractures were individually shorter. We then calculated an effective A√k (fracture surface area × permeability^0.5) based on history-matched production profiles. Despite the significant differences in fracture geometry and count, both spacing scenarios yielded nearly identical A√k values. Despite similar A√k values between the two designs, the standard spacing configuration delivered superior oil production, underscoring the importance of fracture geometry particularly height in high-permeability formations.

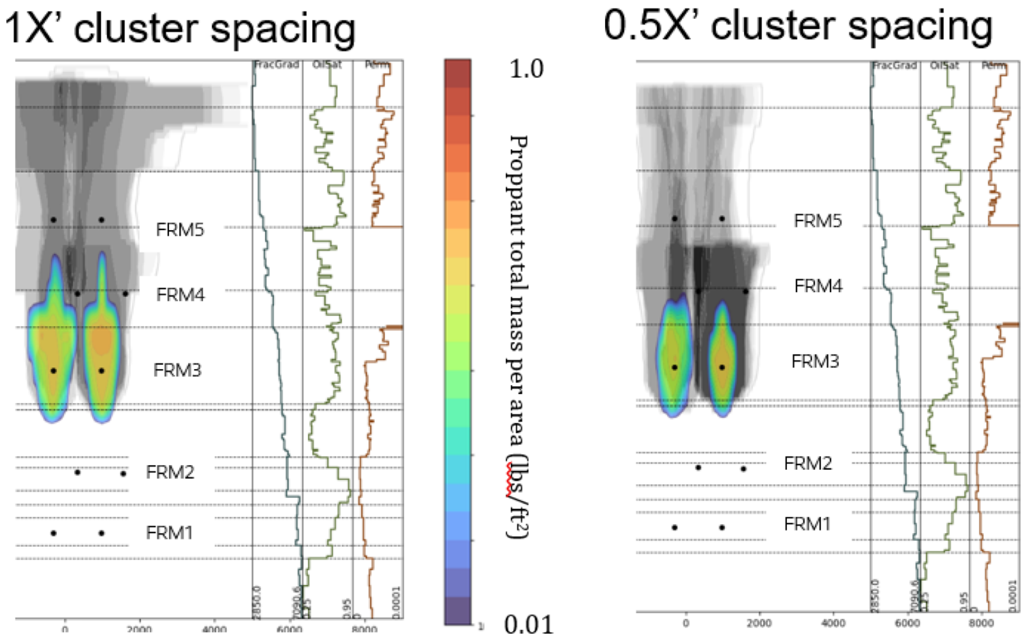


Figure 14—Comparison of propped fracture geometry under standard and reduced cluster-spacing designs in FRM3.

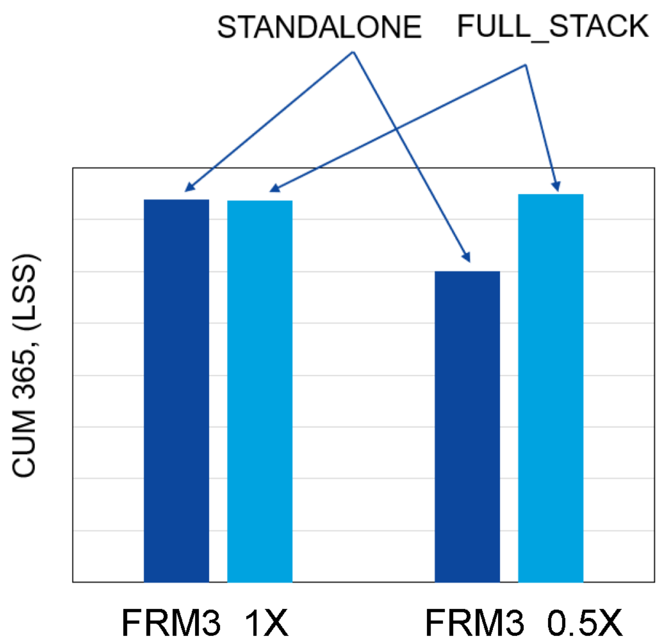


Figure 15—Comparison of one year cumulative production (CUM 365)in FRM3 in standalone vs full-stack

Notably, in the context of the highly permeable FRM3 zone, the simulation suggests that taller fractures are more desirable than simply increasing fracture count. The greater vertical extent enables better access to high-quality rock, which appears to be a more effective strategy for enhancing production in this specific setting.

When these bench-level insights are considered within the context of the full stack (FRM3-FRM5 co-developed), a new dynamic emerges. By day 365, the 0.5X design in FRMM3 not only catches up to but slightly surpasses the 1X case (Fig. 15), as the closer FRM3 fractures interconnect with hydraulic fractures from the FRM4 bench, enabling more efficient drainage of that zone.

However, when the forecast is extended to 40 years, the 1X design ultimately outperforms the 0.5X case by approximately 30% in cumulative oil production. In formations characterized by extensive high-permeability zones, creating taller fractures with greater lateral reach proves more beneficial over the long term than simply increasing fracture count. Therefore, while tighter spacing can accelerate early production through enhanced inter-bench connectivity, maximizing ultimate recovery in these settings depends on leveraging both fracture height and extent to access the largest volume of permeable rock.

Fig. 16 compares the total shallow flow-unit cumulative 365-day production for the standard (1X) and reduced (0.5X) spacing designs. The results clearly show that the standard design delivers superior production performance over the first year.

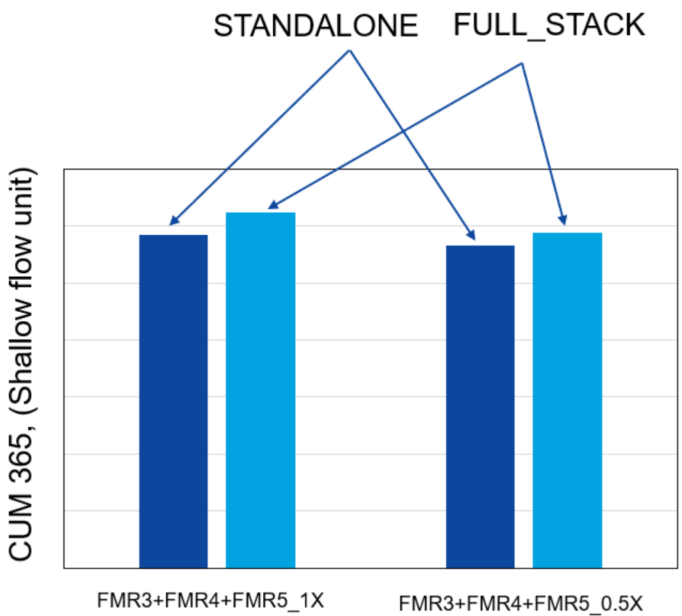


Figure 16—Comparison of one year cumulative production (CUM 365) in the shallow flow unit in standalone vs full-stack

Although one might expect the taller fractures in the standard design to create more inter-bench overlap, they still outperform the reduced-spacing case. This highlights the advantage of generating larger, vertically extensive fractures in a full-stack development context. Taller fractures not only access more high-permeability rock but also enhance inter-bench connectivity, allowing each bench to contribute more effectively to draining the high-quality FRM4layer.

This behavior underscores the superiority of the 1X design over the 0.5X configuration for total flow-unit performance, particularly in formations where vertical connectivity and access to high-permeability zones are critical.

These findings reveal that the impact of reduced versus standard cluster spacing is far less intuitive than initially assumed. It was only through an integrated approach combining analytical modeling, full-physics simulation, and both near and far field diagnostics that we were able to uncover the nuanced trade-offs involved.

An optimal completion design may therefore require moving beyond a uniform application of either reduced or standard cluster spacing across the entire stack. Instead, a hybrid strategy employing reduced cluster spacing in the deeper flow units (FRM1 + FRM2) and standard spacing in the shallower units (FRM3 + FRM4 + FRM5) could capture the advantages of both approaches. Naturally, this recommendation is specific to the current pad density; any changes in well spacing or cluster count could alter these trade-offs and warrant a fresh evaluation.

Conclusions

This study presents an integrated workflow to evaluate the impact of reduced cluster spacing in an unconventional Permian Basin pilot. By combining analytical stress-shadow modeling, near and far field diagnostics, and fully coupled physics-based simulation, we developed a robust framework that provides quantitative insight into the efficiency gains achieved by halving cluster spacing.

Blind test validation on PAD2 confirmed the model's ability to accurately forecast flow unit performance without additional tuning, underscoring its predictive robustness. Near wellbore fiber optic data and perforation erosion measurements, along with standalone time-lapsed geochemistry, further corroborated both the simulation results and analytical insights.

In the deeper flow unit (FRM1 + FRM2), halving cluster spacing proved beneficial: the denser fracture network reduced inter-bench overlap and led to higher cumulative oil production. In contrast, the shallower flow units (FRM3 + FRM4 + FRM5), characterized by significantly higher permeability, responded better to standard spacing. In this case, larger and taller fractures were more effective at accessing far-field, high-permeability rock, ultimately delivering greater recovery and outweighing the benefits of increased fracture count.

These findings demonstrate that optimal cluster spacing is not universally tighter or wider but rather depends on the specifics of reservoir quality and production flow unit characteristics. Under the current well density, a hybrid completion strategy applying reduced clusters spacing in the deeper flow unit and standard spacing in the shallow eremerges as a promising design. As spacing and pad configurations evolve, this integrated methodology should be reapplied to support data-driven completion optimization.

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